

EXPERIENCE



Delft University of Technology

Teaching Assistant

Sep 2023 – Jun 2024 · Part-time

Part-time teaching assistant for multiple first-year BSc Aerospace Engineering courses, supporting tutorials, grading, and student problem-solving.

EDUCATION



University of Oxford Merit

MSc Mathematical Modelling & Scientific Computing

2024 – 2025 · Oxford, United Kingdom

DISSERTATION Distinction · Grade: 71%

Energy and enstrophy preserving discretisations of the incompressible Navier–Stokes equations  with Patrick Farrell and Boris Andrews

extended into a paper, in review at FoCM



University of British Columbia Grade: 85.7%

Minor in Astrophysics & Astronomy

2022 – 2023 · Vancouver, Canada



Delft University of Technology First Class Honours · Grade: 83.4%

BSc Aerospace Engineering

2021 – 2024 · Delft, Netherlands

DISSERTATION Grade: 80%

Regenerative Wind Farming

Supervised by Carlos Simão Ferreira and Uwe Fechner

Bridging Program in Applied Mathematics

2023 – 2024 · Delft, Netherlands

The bridging program in Applied Mathematics at TU Delft is designed for engineering graduates who are to pursue advanced mathematical studies. 

RESEARCH INTERESTS

Structure-preserving/compatible numerical methods for PDEs & ODEs, Conservation & dissipation structures, Global & local energy estimates & conservation laws.

Finite element theory, Finite element exterior calculus (FEEC), Domain decomposition

Computational fluid dynamics, Under-resolved & high-Reynolds-number flows

PUBLICATIONS & PREPRINTS

Preprints

Sep 2026 **Strongly enstrophy–stable integrators for the incompressible Navier–Stokes equations**, with Boris Andrews and Patrick Farrell

◦ In review: *Foundations of Computational Mathematics (FoCM)*

Other works

Oct 2025 **Energy and enstrophy preserving discretisations of the incompressible Navier–Stokes equations**, Supervised by Patrick Farrell

◦ Note: *MSc dissertation (Oxford repository)*

¹Referees: Patrick Farrell, University of Oxford   · Bart Ripperda, University of Toronto   · Kathryn Gillow, University of Oxford  

PROJECTS

- 2023 **On the relativistic magnetized RT instability: An application to**
/24 **magnetically arrested accretion**, with [Bart Ripperda](#) and [Sean Ressler](#)
Through numerical relativistic fluid simulations of Rayleigh–Taylor instabilities, we demonstrated that a dense outer accretion disk penetrates the inner, dilute magnetosphere, facilitating gas particle movement to the event horizon.
- 2023 **Mimetic Spectral Methods with Bundle-Valued Differential Forms**,
/24 with [Marc Gerritsma](#)
I researched the use of bundle-valued forms in Mimetic spectral element methods to preserve the key underlying properties of a PDE in a discrete setting.
- 2023 **Multi-fidelity optimization using GEP for OpenFOAM CFD simula-**
/24 **tions**, with [Richard Dwight](#) and [Renzhi Tian](#)
This project aimed to reduce the simulation time of GEP-generated turbulence models. This work combined symbolic, data-driven machine learning techniques with model selection.

SEMINARS & CONFERENCES

- 2025* **ECCOMAS / MFET**, RWTH Aachen University, **ACOMEN**, Ghent University, **Numerical Analysis Group**, University of Oxford
- 2026* **ESI Programme on Differential Complexes**, University of Vienna, **SciCADE**, University of Edinburgh, **Workshop on Finite Element Tensor Calculus**, Tsinghua University

Minisymposium Presentations*

- 2025 **Firedrake**, University of Leeds
- 2026 **SIAM UKIE '26**, University of Leicester

AWARDS

- Jun **SIAM UKIE '26 Travel Grant**
2026 Awarded to present a poster (available upon request).
- Sep **Barbinder Watson Trust Fund**
2025 Travel grant from St Hugh's College, Oxford, to present a talk at Firedrake '25.
- Feb **Minor Abroad Scholarship**
2023 ~£12 000 awarded to only three individuals to pursue a minor abroad at UBC.
- Feb **STIR Fund**
2023 €400 to support study or research abroad outside the EU/EEA/EFTA.

TRANSLATIONS

- 2026 **Medieval Finitism** by Mohammad Saleh Zarepour
Co-translator: [Kaveh Lajevardi](#)

LANGUAGES

SPOKEN: German (native), Persian (native), English (fluent) · **PROGRAMMING:** Python (Firedrake), Fortran (Tristan-MP, BHAC), C++ (Athena++, H-AMR), R

* I did not attend these; Boris Andrews presented our joint work, which began as my dissertation and was later extended into our joint paper (see Publications & Preprints).

* Presented by me.